

come together

B1 [4]

2 (a) (i) idea of heat lost (by oil) = heat gained (by thermometer)

C1

$$32 \times 1.4 \times (54 - t) = 12 \times 0.18 \times (t - 19)$$

C1

$$t = 52.4^{\circ}\text{C}$$

A1 [3]

(ii) *either* ratio (= $1.6/54$) = 0.030 *or* (= $1.6/327$) = 0.0049

A1 [1]

(b) thermistor thermometer (allow 'resistance thermometer')

B1

because small mass/thermal capacity

B1 [2]

(c) boiling point temperature is constant

M1

further comment

e.g. heating of bulb would affect only rate of boiling

A1 [2]

2 (a) use of $a = \frac{1}{2}at^2$ clear

C1